8.8 Bootstrapping

When one can assume that a population is normally distributed, using confidence interval estimation methods for population parameters, such as the confidence interval estimation of the population mean, is a valid approach to analyzing data. However, when one cannot assume that a population is normally distributed, one uses other methods such as **bootstrapping**, which uses repeated resampling *with* replacement from an initial sample. Because bootstrap methods are based on the original sample, these methods make no assumption concerning the underlying population distribution.

For example, to estimate the population mean using bootstrapping:

1. Select a random sample of size n *without replacement* from a population of size N.
2. Resample the initial sample by selecting n values *with replacement* from the n values in the initial sample.
3. Compute $\overline{X}$ from this resample.
4. Repeat steps 2 and 3 m different times.
5. Construct the resampling distribution of $\overline{X}$.
6. Construct an ordered array of the entire set of resampled means.
7. Find the value that cuts off the smallest $\alpha/2 \times 100\%$ and the value that cuts off the largest $\alpha/2 \times 100\%$. These values provide the lower and upper limits of the bootstrap confidence interval estimate of the population mean.

To use bootstrapping estimation procedures, one typically selects a very large number of resamples (in the thousands) and use statistical software such as JMP or Minitab. To illustrate the bootstrap confidence interval estimate for the population mean, return to Example 8.3 on page 254 that developed a confidence interval estimate of the population mean processing time of applications for life insurance. The confidence interval estimate assumed that the population processing time was normally distributed. Suppose a second insurance company did a similar study of 27 policies that showed that the processing time was right skewed. For this second study, the validity of the confidence interval developed is in doubt.

Table 8.3 shows an initial bootstrap resample for that second study sample of 27 processing times.

TABLE 8.3
Ordered array of first resample of $n = 27$ from a sample of 27 processing times

16	16	16	17	17	17	17	17	19	22	28	31	31	48
51	56	56	60	60	64	64	64	69	**73**	**73**	90	92	

This initial bootstrap resample omits some values (18, 45, 50, 63, and 91) that appear in the initial sample (not shown). Table 8.4 shows the ordered array of the sample means for 100 resamples of the processing times (step 6).

TABLE 8.4
Ordered array of the sample means for 100 resamples of the processing times

31.5926	33.9259	35.4074	36.5185	36.6296	36.9630	37.0370	37.0741
37.1481	37.3704	37.9259	38.1111	38.1481	38.2222	38.2963	38.7407
38.8148	38.8519	38.8889	39.0000	39.1852	39.3333	39.3704	39.6667
40.1481	40.5185	40.6296	40.9259	40.9630	41.2593	41.2963	41.7037
41.8889	42.0741	42.1111	42.1852	42.8519	43.0741	43.1852	43.3704
43.4444	43.7037	43.8148	43.8519	43.8519	43.9259	43.9630	44.1481
44.4074	44.5556	44.7778	45.0000	45.4444	45.5185	45.5556	45.6667
45.7407	45.8519	45.9630	45.9630	46.0000	46.1111	46.2963	46.2963
46.3333	46.3333	46.4815	46.6667	46.7407	46.9630	47.0741	47.2222
47.2963	47.3704	47.4815	47.4815	47.5556	47.6667	47.8519	48.5185
48.8889	49.0000	49.2222	49.4444	49.4815	49.4815	49.6296	49.6296
49.7407	50.2963	50.4074	50.5926	50.9259	51.4074	51.4815	51.5926
51.9259	52.3704	53.4074	54.3333				

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To construct a 90% bootstrap confidence interval, you need to determine the smallest 5% and the largest 5% of the resample means. Since 100 resamples were selected, you need to identify the $0.05 \times 100\% = 5$th smallest value and the $0.05 \times 100\% = 5$th largest value. From Table 8.4, the fifth smallest value is 36.6296 and the fifth largest value is 51.5926. Therefore, the 90% bootstrap confidence interval estimate of the population mean processing time is 36.6296 to 51.5926.

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